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Associations between racial bias and political identity in American cities often diverge from common narratives

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Attitudes about race are often conflated with political identity. However, existing data from cities frequently contradict common narratives about race and politics. To bring more nuance to the discussion of race and political identity, we need an interdisciplinary urban science of political identity formation that accounts for the dynamics of local culture.

Justice and equality have been guiding visions in the American political project. However, a troubled and violent history on matters of race has created a reality where racial inequalities persist in several sectors: healthcare, education, criminal justice, and politics, being a few. Along with the presence of policies that were intentionally implemented to disproportionately affect certain populations, two decades of scientific research suggest that part of the reason for the persistence of racial inequality is the existence of implicit racial biases—attitudes and beliefs about race that operate without conscious awareness or intention[1]. Implicit racial biases occupy an increasingly large place in discussions of racial inequality and political identity in America. In recent years, social psychologists have recognized that these biases are driven by a combination of large-scale cultural factors *and* individual differences in experiences[2]. As a result, a surprising amount of geographic variation in implicit racial biases has been revealed. Here, we focus on data from U.S. cities and highlight that this geographic variation often does not match popular narratives around voting patterns. Specifically, in several U.S. cities, the data contradict the idea that Democratic voters have lower levels of racial biases and Republican voters have higher levels of racial biases. We argue that this mismatch necessitates a reevaluation of discussions around race and political identity formation to include more nuance and local context.

Narratives around race and political identity

Typical contemporary narratives about American political identities associate Democratic voters with tolerance and Republican voters with bias. This framing is commonly used by politicians, journalists, and academic researchers. For example, a recent study[3] demonstrated that people view Black Americans as prototypical Democrats, leading individuals to dramatically overestimate the percentage of Black members in the Democratic Party. These conversations have been especially prevalent in early reflections on the 2024 presidential election, which was defined by counterintuitive patterns in voting by race and ethnicity[4].

However, framing discussions in this way — in the form of general questions about entire populations — does not facilitate discussion or research around variation: How common are these narratives? How much geographical variation is there? How much does local cultural history influence both racial attitudes and the formation of political identities? The pervasiveness of this population-level frame may come from a desire to tell a simple, clear story about American political identity or from the fact that most behavioral scientists prefer to use agreed-upon methods from their respective fields, which limit the types of mechanisms that they can study. The problem with this approach is that explanations of human behavior often require quantitative and qualitative methods that integrate evidence and theories across time and spatial scales, from individual psychology to societal cultural trends.

Although the same broad framing is present in existing research on implicit attitudes about race and the formation of political identity[5,6], psychologists have more recently realized that there are signatures of local environmental influence in implicit racial bias data[2]. As a result, the finding that individuals with higher implicit racial bias levels are less likely to vote for Democratic candidates[5], and its implied inverse, clearly need to be revisited.

Cities facilitate systematic variation in social context

One principled way to revisit the relationship between racial attitudes and politics is to examine what is left over (i.e., the residuals) after accounting for structural factors that can shape implicit biases. For example, a recent interdisciplinary model integrated ideas, empirical results, and mathematical models from psychology, neuroscience, sociology, and urban science to trace individual-level mechanisms of implicit biases across spatial scales, revealing their consequences for entire urban areas[2]. In that model, city population size, racial diversity, and residential segregation structurally shape implicit biases via their impact on the mixing of people from different racial and ethnic groups. More specifically, this model starts from the well-supported premise that implicit biases result from a lack of exposure to people from different racial and ethnic groups. It then accounts for expected exposure rates as a function of city population size (which influences overall social contact rates), and diversity and residential segregation (which influence the probability of inter-group contact). As a result, the residuals from fitting this model to data capture the variation in implicit biases not attributable to these factors and can be interpreted, in part, as reflecting cultures of (implicit) tolerance or bias. For example, a city that is large, diverse, and not very segregated should have low levels of implicit bias. If, instead, that city has high levels of implicit bias, the difference between expectation and reality can be interpreted as reflecting a culture of implicit bias.

There are two related reasons that we center our argument on this particular model of implicit biases in cities, rather than other spatial units. First, social psychologists increasingly recognize the important role of social context in shaping implicit biases[2]. As a result, we strive to work with spatial units that allow for systematic comparisons across social contexts. Second,

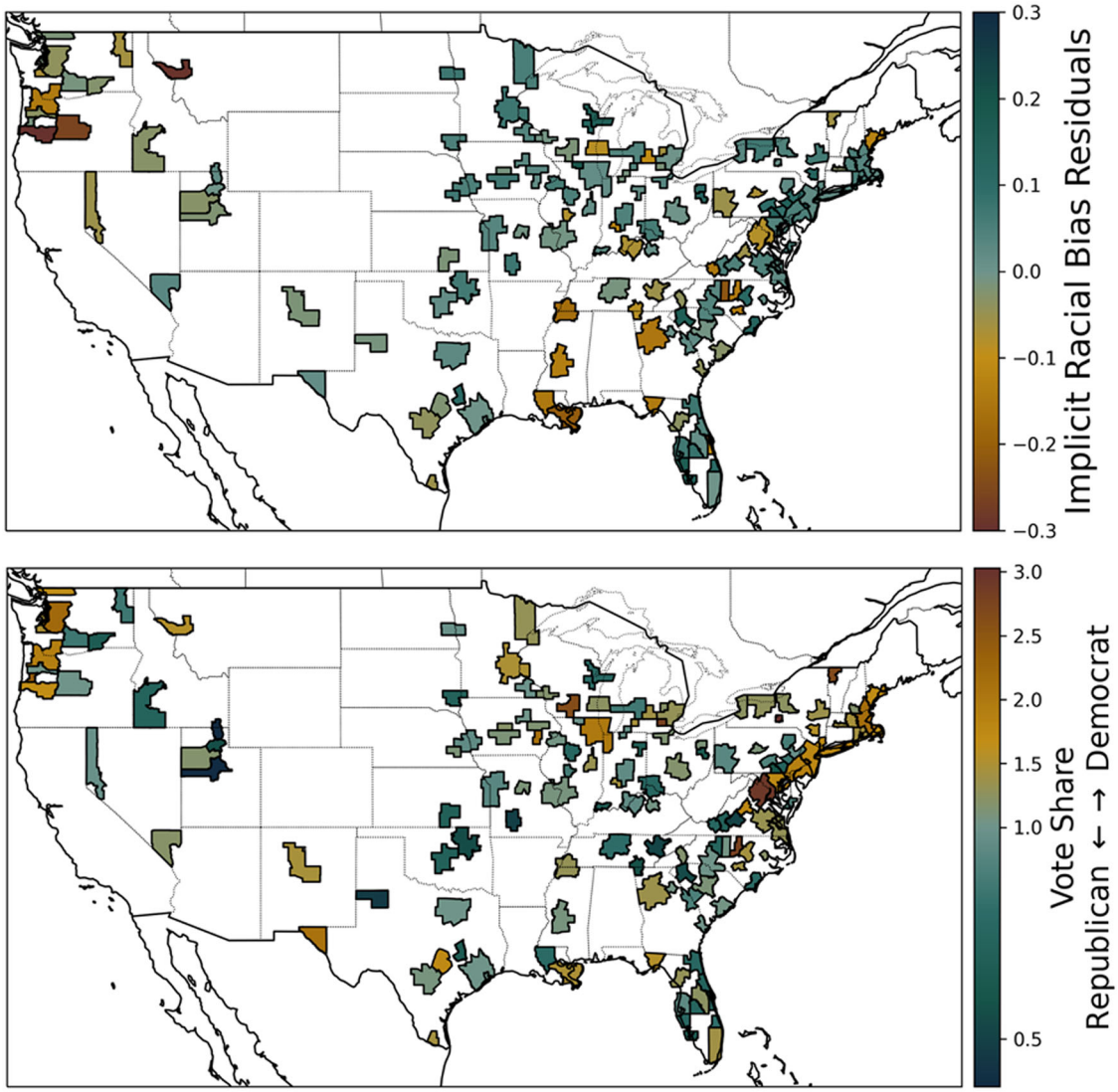


Fig. 1 | Greater metropolitan area implicit racial bias residuals[2] and Presidential election vote ratios[12] in 146 US cities from 2020. Top: Residuals from a mathematical model that accounts for how city population size, diversity, and segregation influence implicit racial biases via mixing between individuals from different racial and ethnic groups. Bottom: the ratio of votes for the Democratic candidate to votes for the Republican candidate in the 2020 presidential election. Both panels use cities defined as core-based statistical areas by the US Census.

the past twenty years of urban science have demonstrated that cities, in their many modern and ancient forms, are surprisingly consistent in the ways in which they shape human behavior and, more generally, social contexts[7]. This is in contrast to other types of human societies that exist in, e.g., rural areas, where long-term cultural trajectories often drive social contexts. Thus, while many rural areas differ in different ways, all cities force their inhabitants to travel over the same road networks and use the same public transit regardless of their religion, ethnicity, political beliefs, or lifestyle. This creates predictable social mixing among city inhabitants and provides a way to systematically study variations in social context that are difficult, if not impossible, to identify in the less structured physical environments of rural areas.

In addition to these theoretical motivations to focus on cities, other spatial units at which implicit bias and voting data are available are often not comparable across places. For example, Santa Fe County covers the entire Santa Fe, New Mexico metropolitan area, while New York County covers only the island of Manhattan, a small fraction of the greater New York metropolitan area. Similarly, while Wyoming has two metropolitan areas — the largest of which has approximately 100,000 inhabitants — Texas has 18 metropolitan areas with more than 100,000 inhabitants. Although these spatial units (county and state) are administratively relevant, they do not capture comparable social contexts across places. In contrast, the most commonly used definition of city boundaries is based on a threshold of residents who commute to a city center or commute from the city center to an outlying area. In the U.S. these are metropolitan statistical areas (MSAs, the definition we rely on here), and in the E.U. they are functional urban areas (FUAs). Because these definitions capture the spatially extended socio-economic networks that cities facilitate[8], they are comparable between

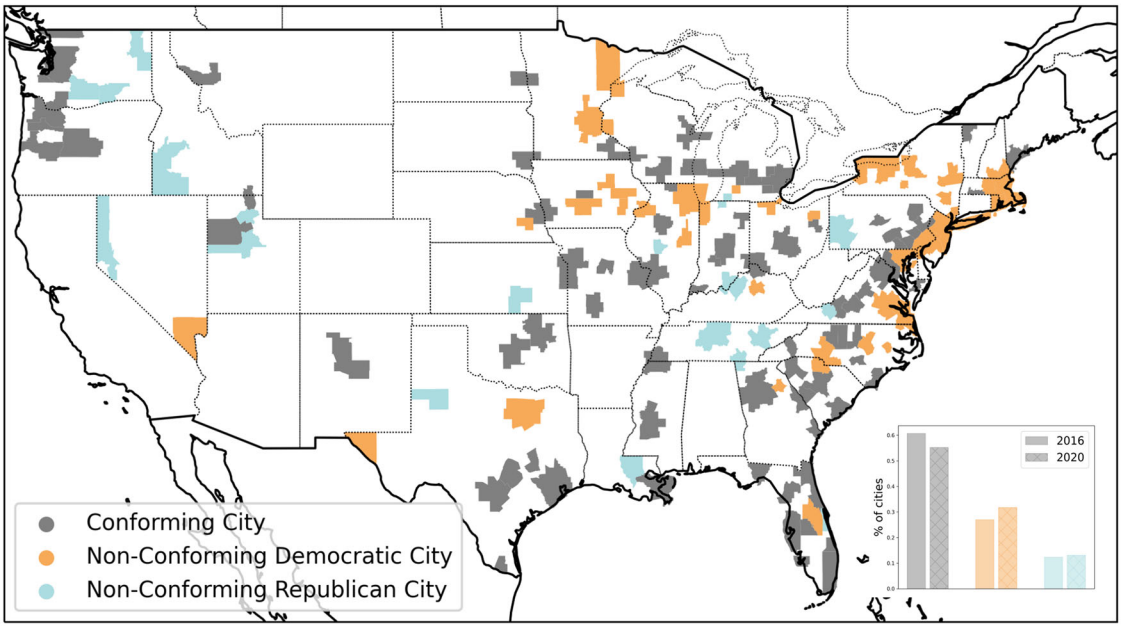


Fig. 2 | Cities that do not conform to the typical narrative around racial attitudes and political identity formation (2020 election). Non-conforming cities are defined as those that voted for the Democratic candidate (vote ratio greater than one in Fig. 1 bottom) but have higher than expected bias levels (positive residuals in Fig. 1 top) or those that voted for the Republican candidate (vote ratio less than one) but have lower than expected bias levels (negative residuals). Inset: the percentage of cities in each category for 2020 and 2016.

places, even while they include city centers, suburban communities, and sometimes rural areas.

Where do urban racial biases and voting patterns (Mis)align?

In Fig. 1, we present maps of the residual bias levels from the mathematical model of urban implicit biases[2] and the ratio of Democratic to Republican votes for the 2020 presidential election in cities for which we have implicit bias data. There are clear geographic patterns in these maps. For example, many cities in the Northeast and Pacific Northwest voted for the Democratic candidate, while cities in the South and Midwest tended to vote for the Republican candidate (Fig. 1 bottom). In contrast, the residual bias map (Fig. 1 top) is less conforming to the common narrative around racial attitudes and politics. There are cities, for example, in the Pacific Northwest and parts of the Midwest and Texas, which conform to the Democratic-low-bias, Republican-high-bias narrative. However, many cities do not conform to this narrative. For example, cities in the Northeast, except for Portland, ME, and Washington, D.C., voted for the Democratic candidate but had positive residuals indicating higher implicit racial biases than expected. In contrast, cities in Tennessee tended to vote for the Republican candidate but had much lower implicit racial biases than expected (as indicated by negative residuals).

In Fig. 2, we highlight all 58 cities that do not conform to the common narrative out of the 146 cities in the sample with implicit racial bias data. We also list a sample of conforming and non-conforming cities in Box 1. Note that in 2016, the proportions of conforming, non-conforming Democratic voting and non-conforming Republican voting cities were similar, even though the 2020 and 2016 elections resulted in different political parties being elected to office (Fig. 2 inset, Supplementary Figs. 1, 2). Across both years, many cities do not conform to the common narrative, even though a majority of cities do conform to the common narrative.

It is tempting to start building stories about the geographic distribution of nonconforming cities. For example, one potential argument is that the nonconforming Democratic voting cities tend to be more populous, and so these patterns might be explained by cities where strong labor movements took place and continue to influence politics to this day. However, many smaller cities, particularly in the Midwest and South Atlantic, are non-conforming and vote for Democratic candidates. In addition, the non-conforming cities are spread throughout regions in the US with different histories of political identity, racial attitudes, and racial and ethnic diversity. This suggests that to understand maps like Fig. 2, we need models that combine the general principles by which biases[2,5] and political identities[6] form with the effects of local cultural histories.

Towards integrated, data-driven narratives of political identity

Integrating data on implicit racial bias and voting patterns across 146 US cities reveals a complex relationship between implicit racial bias and political identity. The picture that emerges does not conform to common narratives: Many American cities that vote Democrat have high implicit racial bias scores, and several that vote Republican have low implicit racial bias scores. This result refines our picture of the American political landscape and highlights how the relationship between racial bias and political identity is one example of many in the social sciences that require multiple lenses to understand.

And while the present analysis focuses on the relationship between implicit bias and political identity in cities, we offer that this problem is reflective of many such complex social systems, where assumptions, public narratives, stereotypes, or intuition may lead us astray. For example, researchers have used an interdisciplinary suite of methods and data to understand the peculiar properties of academic prestige in hiring and

Box 1 | A sample of conforming and non-conforming cities (2020 election)

	Democratic Voting Cities	Republican Voting Cities
Conforming Cities (Democrat-Low Bias, Republican-High Bias)	Eugene-Springfield, OR Missoula, MT Bend, OR Greensboro-High Point, NC New Orleans-Metairie, LA	Reading, PA Green Bay, WI Greenville, SC North Port, FL Rockford, IL
Non-Conforming Cities (Democrat-High Bias, Republican-Low Bias)	Davenport, IA-IL Albany, NY Des Moines, IA Rochester, NY Lincoln, NE Minneapolis, MN-WI Trenton-Princeton, NJ Akron, OH Raleigh-Cary, NC Kingston, NY	Blacksburg, VA Baton Rouge, LA Palm Bay, FL Chattanooga, TN-GA Springfield, IL Louisville, KY-IN Spokane, WA Knoxville, TN Pittsburgh, PA Reno, NV

promotion[9] and racial disparities in solitary confinement[10]. These studies have revealed previously unrecognized sources of inequality and demonstrate the power of multiple lenses in understanding the forces operating in complex social systems. However, they focus on the specific contexts of academic and carceral institutions. Nonetheless, racial attitudes and political identities are relevant to the functioning of many types of social institutions and broadly to the functioning of entire societies.

We propose that the complex systems science interdisciplinary toolkit is already well suited to enhance our understanding of the relationship between racial attitudes and political identity in a less specific context. Specifically, we assert that this set of tools can help curb the proliferation of theories that are loosely related but difficult to compare or reconcile, making synthesis challenging. This proliferation has occurred because existing quantitative investigations lack coherence between the constructs used to measure racial bias and political attitudes and because there is a disconnect between verbally specified theories (in contrast to mathematically specified) and statistical models[11]. To overcome these barriers to synthesis, quantitative researchers should focus on developing context-specific mathematical theories of bias and political identity formation and change. At each stage, these mathematical models can be directly mapped onto statistical models so that the various ways in which researchers construct empirical measures of racial attitudes and political identity can be directly compared to each other, and with predictions generated from theory.

This connection between measurement and theory will help to enumerate the basic mechanisms of racial attitude and political identity formation (for example, that implicit racial biases form, in part, through expertise-based learning via interactions with out-group individuals[2]). These mechanisms should also necessarily draw inspiration from the social sciences and humanities and, in turn, inform new qualitative studies, including ethnographic interviews, content analysis, case studies, and case comparisons, to better understand how local cultural histories might give rise to observations like those shown in Fig. 2.

There are new models to build and new data to collect. However, we propose that the quantitative and qualitative methods described above, when combined, can reveal how the complex shape of American history manifests in the formation of political identities in cities, and beyond. What remains missing are individual and institutional commitments to this research.

Data availability

All data used to make the figures in this manuscript are publicly available. Implicit racial bias residuals[2] are available at https://github.com/enlberman/implicit biases cities/ and county-level Presidential election returns[12] are available at https://electionlab.mit.edu/data.

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The authors declare no competing interests.

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